

Supplement: Diagnosing Hybrid Attention/Mamba Cache Over-Reservation

Anonymous Workshop Submission

1 Artifact Map

All paths are relative to the repository root. The supplement preserves the path-heavy reproducibility material omitted from the main workshop paper.

Artifact	Repository path
Diagnostic report	research/avmp/v2/results/ vllm-runtime-cache-diagnostic-cli/report.json
Diagnostic summary	research/avmp/v2/results/ vllm-runtime-cache-diagnostic-cli/summary.md
Diagnostic manifest	research/avmp/v2/results/ vllm-runtime-cache-diagnostic-cli/manifest.json
Planner replay output	research/avmp/v2/results/vllm-planner-stage-observation/ translated_planner_stage_config.json
Planner-stage advisory diff	research/avmp/v2/results/vllm-planner-stage-advisory-diff/ diff_report.json
Group-level advisory diff	research/avmp/v2/results/vllm-planner-stage-advisory-diff/ group_level_diff.json
Path C matrix	research/avmp/v2/evaluation/matrix_table.md and research/avmp/v2/evaluation/matrix_table.csv
Planner-only pack	benchmarks/results/rtx3060/planner-comparison/
Claim boundary	research/avmp/v2/evaluation/claim_summary.md

2 Exact Byte Tables

The main paper reports GiB-rounded values for readability. Tables below retain the exact byte values used to construct the main-paper tables and ratio figure.

Len	Util.	Reserved	Useful	Savings
2048	0.6	1,893,335,040	1,092,283,392	801,051,648
2048	0.7	3,185,049,600	1,837,486,080	1,347,563,520
4096	0.6	1,619,066,880	934,055,424	685,011,456
4096	0.7	2,910,781,440	1,679,258,112	1,231,523,328

Table 2: Exact Path C advisory matrix byte values.

3 Reproduction Commands

3.1 Focused Tests

```
UV_CACHE_DIR=/tmp/uv-cache uv run pytest \  
tests/bench/test_planner_comparison.py \
```

Workload	Planner	Useful	Estimated	Ratio	Waste	OOM
short-heavy	padded	2,435,055,616	6,985,613,312	2.868770	0.651418	no
short-heavy	AVMP	2,435,055,616	2,451,046,400	1.006567	0.006524	no
long-heavy	padded	41,967,550,464	165,116,116,992	3.934376	0.745830	yes
long-heavy	AVMP	41,967,550,464	41,983,672,320	1.000384	0.000384	yes
mixed	padded	13,517,422,592	51,311,017,984	3.795917	0.736559	yes
mixed	AVMP	13,517,422,592	13,532,397,568	1.001108	0.001107	yes

Table 3: Exact RTX 3060 planner-only comparison byte values.

```
tests/bench/test_vllm_path_c_advisory_matrix.py \
tests/cli/test_diagnose_vllm.py -q
```

3.2 Planner Pack Regeneration

```
UV_CACHE_DIR=/tmp/uv-cache uv run python \
  benchmarks/scripts/create_planner_comparison_pack.py \
  --output-dir /tmp/cachepawl-planner-comparison-workshop-check \
  --seed 1 --num-requests 128 \
  --gpu-name "NVIDIA GeForce RTX 3060" \
  --gpu-total-bytes 12884901888
```

4 Runtime Contract Boundary

The current evidence resolves planner-stage replay, scheduler and KV-manager structure, request-to-block assignment, worker tensor layout, and attention block-table metadata. Mamba state-index and Mamba state tensor contracts remain blocked. Consequently, controlled substitution remains future work.